CSE 211 · Theory of Computation

Previous-Year Question Analysis & Study Priority

Based on 17 BUET term-final papers (2006-07 → 2023-24, the latest held May 2026), 369 question parts worth 3749 marks in total, each tagged to a topic and mapped to Hakim Sir's and Atif Sir's slides.

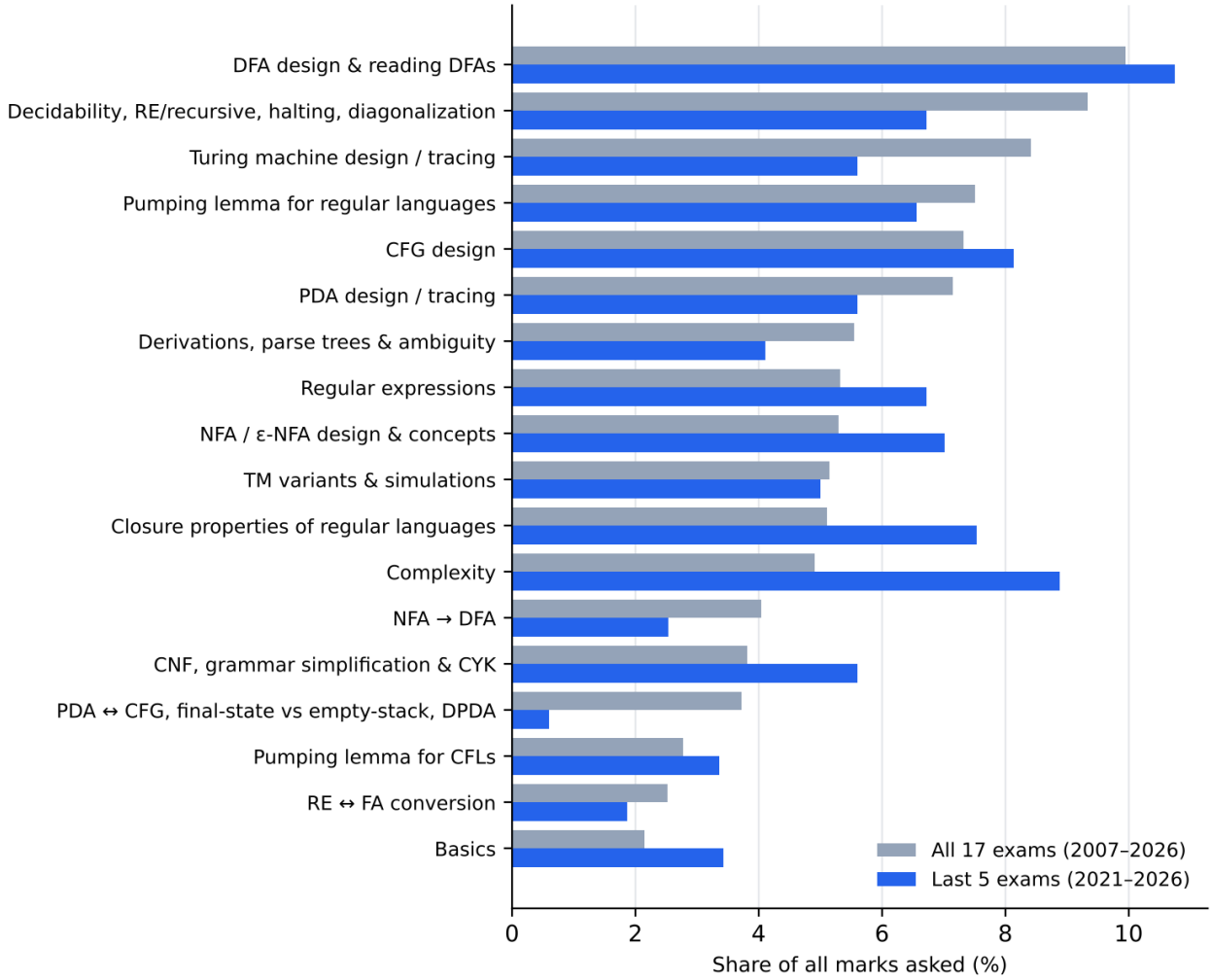
Key findings

- Most-asked topics overall (by marks): DFA design (9.9%), Decidability/undecidability (9.3%), TM design (8.4%), Pumping lemma for regular languages (7.5%), CFG design (7.3%) and PDA design (7.1%). Each of these six appeared in all 17 papers.
- Rising in recent papers: Complexity / NP-completeness / Cook's theorem went from absent before 2019 to 8.9% of recent marks; Closure properties (7.5%) and CNF + CYK (a full 15-mark question in every one of the last 5 papers) also rose.
- Fading: PDA ↔ CFG conversions and final-state vs empty-stack proofs (huge before 2017, almost gone now), formal NFA→DFA proofs, and DFA→RE state elimination.
- Section B has become almost a fixed template since 2019: CFG design · CFG + ambiguity · CNF/CYK · PDA · pumping lemma for CFL · TM design (bitwise op) · multi-track vs multitape / simulation true-false · Diagonalization/Halting · Cook's theorem & PSPACE = NPSPACE. Several are repeated almost word-for-word.
- Study first: DFA design → pumping lemma (regular) → CFG design + ambiguity → CNF/CYK → PDA design → TM design → undecidability → complexity. The full plan is on page 7.

How the analysis was done

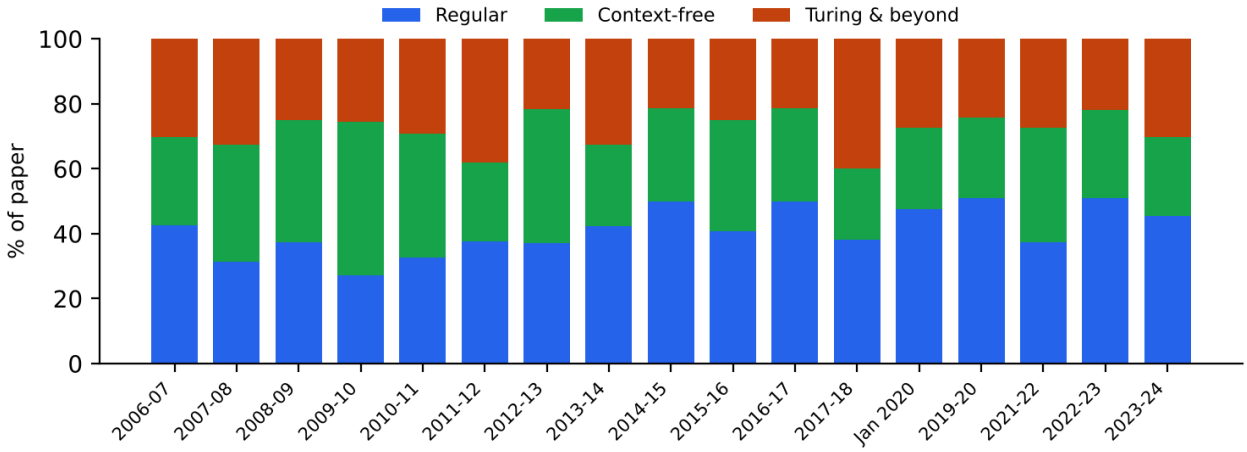
- Every sub-question (a, b, c …) of both sections was tagged with one of 18 topics that follow the slide decks. Questions spanning two topics were split by their marks.
- "Share of marks" = marks for a topic ÷ all marks printed on the papers. Papers have internal choice (e.g. answer 3 of 4, or 7 of 9), so this measures what examiners ask, which is what you need to prepare for.
- "Recent" = the last 5 papers (Jan-2020 term, 2019-20, 2021-22, 2022-23, 2023-24). These follow the current format: 210 marks, Section A with a compulsory Q1 plus 2 of 3, Section B with 7 of 9 fifteen-mark questions.
- Priority score = 40% all-time share + 60% recent share, so current trends count more than 2008 papers.

1. Which topics carry the most marks?



Grey bars show the share across all 17 papers; blue bars show the share in the last 5. Where blue is longer than grey (Complexity, Closure, NFA, CNF), the topic is gaining weight. Where it is shorter (PDA ↔ CFG, NFA→DFA, TM design, Undecidability), it is still asked but takes fewer marks than before.

Unit balance in each paper



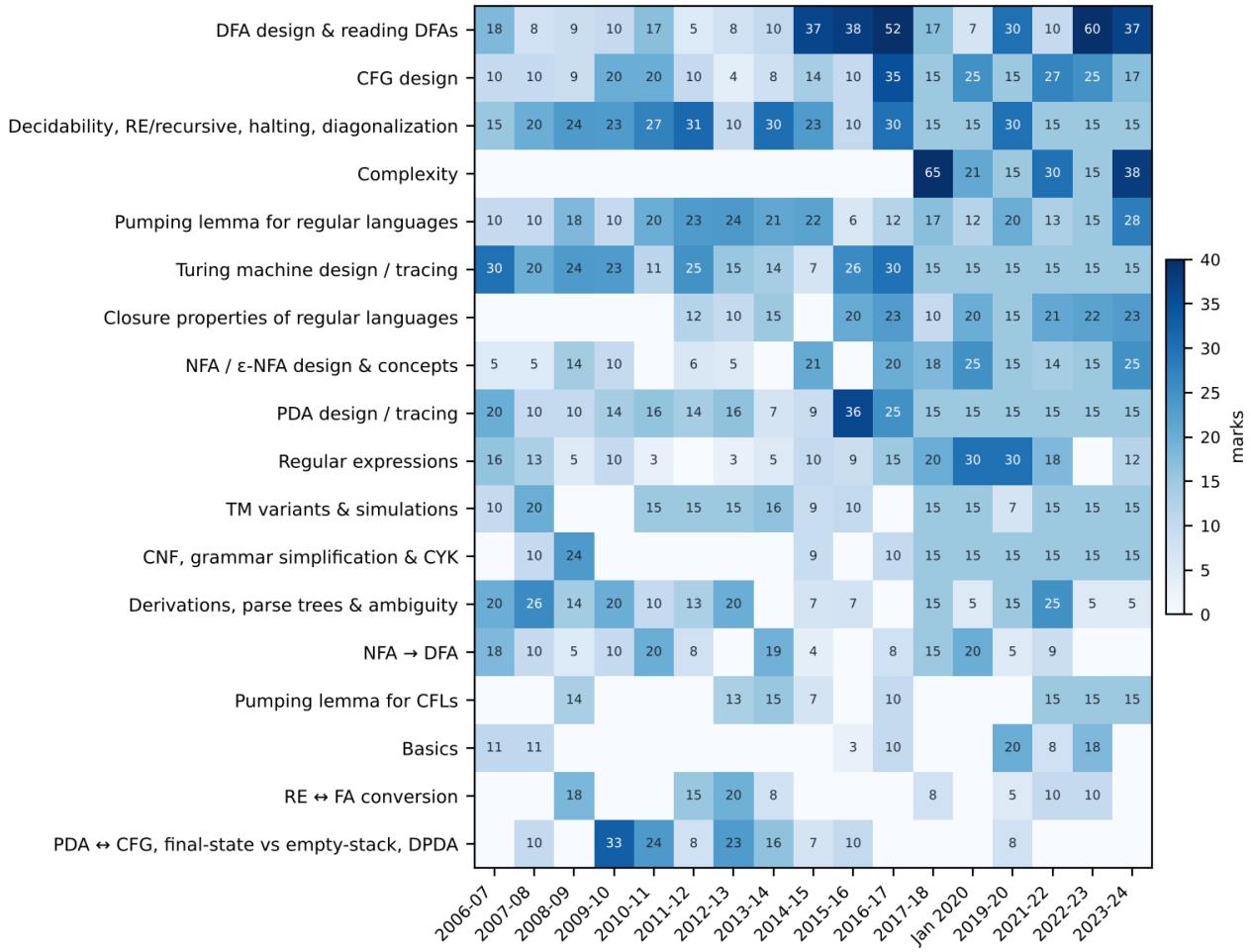
Since 2017 the paper is split about 50/50: Section A covers regular languages (Hakim Sir, lectures 1–6) and Section B covers context-free languages, Turing machines and complexity (Atif Sir, plus Hakim Sir's CFG and PDA lectures). Regular languages alone are about half the paper, so they matter most.

2. Ranked topic table

#	Topic	Slides	All %	Recent %	Parts	Papers	Priority
1	**DFA design & reading DFAs**	Hakim L3	9.9	10.7	33	17	Must
2	**CFG design**	Hakim L7	7.3	8.1	27	17	Must
3	**Decidability, RE/recursive, halting, diagonalization**	Atif L7	9.3	6.7	39	17	Must
4	**Complexity: P, NP, NP-complete, Cook, PSPACE**	Atif L8-L9	4.9	8.9	14	6	Must
5	**Pumping lemma for regular languages**	Hakim L6	7.5	6.6	28	17	Must
6	**Turing machine design / tracing**	Atif L4-L5	8.4	5.6	31	17	Must
7	**Closure properties of regular languages**	Hakim L5	5.1	7.5	17	11	High
8	**NFA / ε-NFA design & concepts**	Hakim L4	5.3	7.0	19	14	High
9	**PDA design / tracing**	Hakim L8, Atif L3	7.1	5.6	24	17	High
10	**Regular expressions (write / describe / simplify)**	Hakim L5	5.3	6.7	21	15	High
11	**TM variants & simulations (multitape, NTM, tracks, computer)**	Atif L6	5.1	5.0	18	14	High
12	**CNF, grammar simplification & CYK**	Atif L2	3.8	5.6	10	10	High
13	**Derivations, parse trees & ambiguity**	Hakim L7	5.5	4.1	24	15	Medium
14	**NFA → DFA (subset construction, ε-closure)**	Hakim L4	4.0	2.5	16	13	Medium
15	**Pumping lemma for CFLs**	Hakim L7	2.8	3.4	9	8	Medium
16	**Basics: languages, ε/∅, role of states, applications**	Hakim L1-L2	2.1	3.4	12	7	Medium
17	**RE ↔ FA conversion (RE→NFA, DFA→RE)**	Hakim L5	2.5	1.9	11	8	Medium
18	**PDA ↔ CFG, final-state vs empty-stack, DPDA**	Hakim L8, Atif L3	3.7	0.6	16	9	Medium

Must = top 6 by priority score: expect at least one full question on each in every paper. High = usually appears, often as a 10–15-mark part. Medium = appears in some years, often as short conceptual parts.

3. Topic × paper heat-map (marks)



Rows are in priority order. A row that is dark across the right-hand side (recent papers) is the strongest signal. CNF/CYK, PDA, TM design, Undecidability and Complexity have been 15 marks every year since 2019.

4. Topic notes: what exactly gets asked

DFA design & reading DFAs [Must]

- Counting / modular DFAs: binary number divisible by 3, 4 or 5; base-3 numbers not divisible by 4; “sum of digits mod 3 or 5 with an R reset symbol” (2014-15, 2019-20, 2022-23 Q1b, a 3-state vs 2-state argument).
- Substring and run constraints: no aa/bb, no substring 110, begins with a but no aab, “at most two runs of a’s of length exactly three” (2023-24), no even run of 0s (2022-23).
- Reading a diagram: “describe informally the language accepted” (2022-23, 20 marks, 5 diagrams) and short conceptual questions (does the number of accept states change the table size?).

Pumping lemma for regular languages [Must]

- Classic proofs: 0^n1^n, prime-length 1^p (asked 4+ times), perfect squares, a^n with $n = k^2$, $0^n1^n0^n$, www, $a^{2n}b^{3n}a^n$.
- “Spot the flaw” questions (Surcharged Skye / B L Perfunctory) appear in 2017-18, Jan-2020 and 2021-22. You must know exactly what you may choose (w) and what the adversary chooses (x, y, z with $|xy| \leq p$, $|y| > 0$).
- Newer conceptual parts: why $|xy| \leq p$ is needed (2023-24), finding the minimum pumping length (2022-23), why the lemma only gives “negative proofs”. Also be ready to state and prove the lemma.

CFG design + derivations / ambiguity [Must]

- Fixed favourites: balanced parentheses, equal numbers of 0s and 1s, “at least three 1s”, odd length with 0 in the middle, palindromes, $a^ib^jc^k$ with $i = j$ or $j = k$, or $i \neq j + k$.
- $\{a^nb^nc^md^m\} \cup \{a^nb^mc^md^n\}$: design the grammar and show it is ambiguous. This appeared in 2008-09, 2012-13, Jan-2020, 2022-23, and a variant in 2023-24.
- Ambiguity: E → E+E | E*E | (E) | a (two parse trees, then remove the ambiguity); S → AB | C … grammar (2019-20 and 2021-22, identical); define inherent ambiguity.

CNF, simplification & CYK [Must]

- Convert to CNF with all steps (ε-productions, unit productions, useless symbols). A → BAB | B | ε, B → 00 | ε has been asked twice (2014-15, 2016-17).
- CYK with S → BZ | CY | ZZ | YY, B → ZA, C → YA, A → ZA | YA | 0 | 1, with only the test string changing: 10011, 01100, 0010, 0100. This is a guaranteed 15 marks if you practise the table.

PDA design / tracing [Must]

- $a^ib^jc^k$ with $i = j$ or $i = k$ (asked in 2013-14, Jan-2020 and 2022-23), $a^nb^mc^md^n$, wcw^R / ww^R, equal numbers of a’s and b’s, a^nb^{2n}, a^nb^m with $n \leq m \leq 2n$.
- ARITH: a PDA for legal arithmetic expressions (2019-20 and 2021-22, identical). Practise tracing IDs for a given input.

Turing machine design [Must]

- Section B always has a binary-tape TM: bitwise XOR / OR / AND / NOT, equality test, two’s complement, binary +1, using multiple tracks and storage in the state.
- Older papers ask for 0^n1^n, $a^nb^nc^n$, wcw^R, w-w, a^(2^n), and unary subtraction. Learn one marking-and-crossing template and one track-based template well.

Decidability / undecidability [Must]

- Diagonalization language L_d is not RE; the universal language; the halting problem is RE but not recursive. These appear every year at 15 marks.
- Give an example from each class (regular, CFL-not-regular, recursive-not-CFL, RE-not-recursive, not-RE); recursive vs RE; L and its complement both RE ⇒ recursive; Rice's theorem (older papers); countability of TMs and of binary sequences.

Complexity (P / NP / PSPACE) [High · rising]

- Cook's theorem: how every NP problem reduces to SAT, and why a polynomial-time algorithm for an NP-complete problem gives P = NP. Asked in Jan-2020, 2021-22, 2022-23 and 2023-24.
- Venn diagram / relationships among P, NP, NP-complete, PSPACE, NPSPACE, EXPTIME and recursive/RE; PSPACE = NPSPACE (Savitch). Also NP-completeness proofs (3-colourability, subgraph isomorphism) and certificates (Dominating Set).

TM variants & simulation [High]

- True/false items: a one-tape multi-track TM can simulate a multitape TM; a DTM simulates n steps of a computer in p(n) steps; a computer simulates an NTM in p(n) steps (false as far as is known). These recur almost every year.
- Equivalence proofs: multitape → single tape, NTM → DTM, two-way infinite tape vs semi-infinite tape.

Closure properties & constructions [High · rising]

- Union of two NFAs (write δ for N and explain each case), concatenation, star. The flawed star construction ("just make the start state accepting") appeared in 2016-17 and 2023-24, and the Jan-2020 term asked you to explain each case of the star δ.
- Complement by swapping accept states works for a DFA but not for an NFA: give a counterexample (2015-16, 2017-18, 2021-22, 2022-23).

NFA design, NFA → DFA, regular expressions [High]

- NFA with few states (e.g. $a^n \cup b^m a^k$ with 3 states), "final digit appeared before", "accepts {a}* but becomes {a} when one edge is removed".
- Subset construction from a table or diagram (min/max DFA states from a 5-state NFA). Write REs for count-mod-3, at most three a's, exactly one aaa, and pick the correct RE from options with reasons (2021-22, 2022-23, 2023-24).
- Why is the Kleene star of ∅ equal to {ε}? asked in 2016-17, 2021-22 and 2022-23.

Pumping lemma for CFLs [High]

- $a^n b^n c^n$ is the standard example. "Give a non-CFL and prove it" has been a 15-mark question in 2021-22, 2023-24 and 2022-23 (state and prove the lemma using the CNF parse-tree bound $|w| \leq 2^{n-1}$).

5. The current Section B template

The last 5 papers follow almost the same 9-question pattern (the order shifts slightly from year to year) (answer 7 × 15 marks = 105). This makes Section B the most predictable part of the paper:

Q	Typical content (2019-20 → 2023-24)	Topic
5	CFGs: balanced parentheses / equal 0s and 1s / at least two or three 1s	CFG design
6	Grammar for $a^n b^n c^m d^m \cup a^n b^m c^m d^n$ (or similar) and show ambiguity; or define inherent ambiguity	Ambiguity
7	CYK table on the fixed CNF grammar (or a CNF conversion)	CNF / CYK
8	PDA for $a^i b^j c^k$ (i = j or i = k), $a^n b^m c^m d^n$, or ARITH	PDA
9	Pumping lemma for CFLs (state + prove, or a non-CFL example)	PLC
10	TM for a bitwise op on wcw (XOR / AND / NOT / equality / 2's complement)	TM design
11	True/false: multi-track one-tape simulating multitape; polynomial simulations	TM variants
12	Diagonalization language not RE / Halting problem not recursive	Undecidability
13	Cook's theorem + P vs NP / PSPACE vs NPSPACE relationships	Complexity

Section A (regular languages) is less fixed. Its compulsory Q1 now mixes a short "ToC in the real world" part (password login, public-key cryptography, "what if a problem seems hard?") with a DFA design, a closure construction and a pumping-lemma critique. Practise explaining as well as constructing: many parts say "drawing only will not get full marks" or "explain why each option is right or wrong".

6. Recommended study order

Phase	Study	Why first	Slides
1	DFA design → NFA + subset construction → regular expressions	Foundation for half the paper; DFA design is the single largest topic and appears in all 17 papers	Hakim L3–L5
2	Pumping lemma (regular) + closure-property constructions	Appears in every paper; the flaw-finding and construction-justification parts are now standard in Q1–Q4	Hakim L5–L6
3	CFG design + ambiguity → CNF conversion + CYK	Three of the nine Section B questions; highly repetitive, so practice pays off quickly	Hakim L7, Atif L2
4	PDA design → pumping lemma for CFLs	Two more Section B questions; reuse the same i = j or i = k style languages	Hakim L8, Atif L3
5	TM design (bitwise ops, multiple tracks) → TM variants	One design and one true/false question every year	Atif L4–L6
6	Undecidability: L_d, universal language, halting, language hierarchy	15 marks every year, mostly standard proofs to memorise and understand	Atif L7
7	Complexity: Cook's theorem, P/NP/NPC, PSPACE = NPSPACE	Rising fast; theory-heavy but short, so leave it for the final days	Atif L8–L9
8	Lower priority: PDA ↔ CFG conversions, DFA → RE elimination, Rice's theorem	Mostly older papers; skim if time allows	Atif L3, L7; Hakim L5

If you are short on time: do phases 3, 4 and 6 first. Section B repeats itself so closely that solving the last 5 papers' Section B nearly covers it. Then do phases 1–2, which are bigger but needed for Section A's compulsory Q1.

Practice suggestion: solve the 2023-24, 2022-23 and 2021-22 papers fully under timed conditions. Most questions in the last few papers are reworded versions of each other.